

LINUX LAB-1

Que-2

Que 2: Demonstrate pipes & redirection (>, >>, <, 2>, 2>>, 2>&1, |) with cat, grep, sort, uniq, wc, tee to generate filtered reports.

Ans 2: Linux provides pipes and redirection operators for transferring and storing command output. The pipe (|) sends the output of one command as the input of another command. The > operator redirects output to a file and overwrites its previous content, while >> appends output to an existing file. The < operator provides input from a file. 2> and 2>> redirect standard error to a file, while 2>&1 combines standard error with standard output. The tee command saves output into a file while also passing it to the next command. Commands such as cat, grep, sort, uniq, and wc can be combined using these operators to generate filtered reports.

Result

The pipes and redirection operations were successfully demonstrated using cat, grep, sort, uniq, wc, and tee.

Filtered data, sorted unique records, line counts, and error reports were successfully generated.

```
luffy: /home/LabFiles  x  +  v
fffy:/home/LabFiles# cat > sample_data.txt << 'EOF'
10
na 20
10
20
fffy:/home/LabFiles# cat sample_data.txt | grep apple > apple_report.txt
fffy:/home/LabFiles# cat sample_data.txt | sort | uniq >> apple_report.txt
fffy:/home/LabFiles# wc -l sample_data.txt >> apple_report.txt
fffy:/home/LabFiles# cat sample_data.txt | tee full_copy.txt | wc -l
fffy:/home/LabFiles# ls -l /nonexistent 2> error_log.txt
fffy:/home/LabFiles# cat sample_data.txt 1> out.txt 2> err.txt
fffy:/home/LabFiles# cat sample_data.txt 2>&1 | tee combined.txt > /dev/null
fffy:/home/LabFiles# ls -l apple_report.txt full_copy.txt error_log.txt out.txt err.txt combined.txt
r-- 1 root root 64 Sep 19 17:09 apple_report.txt
r-- 1 root root 37 Sep 19 17:09 combined.txt
r-- 1 root root  0 Sep 19 17:09 err.txt
r-- 1 root root 60 Sep 19 17:09 error_log.txt
r-- 1 root root 37 Sep 19 17:09 full_copy.txt
r-- 1 root root 37 Sep 19 17:09 out.txt
fffy:/home/LabFiles# |
```